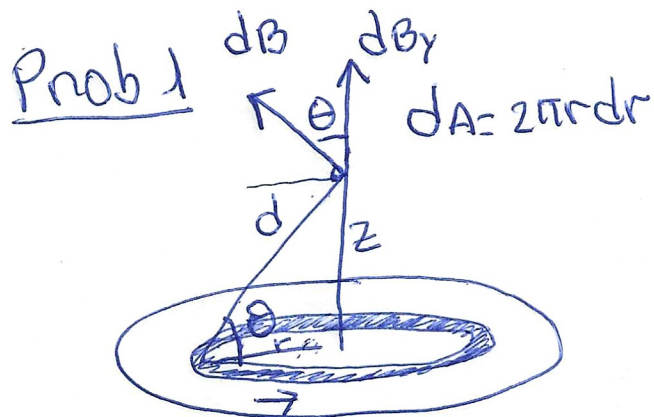


Examen final Física III

Soluciones



$$\sigma = \frac{dq}{2\pi r dr} \quad dq = \sigma 2\pi r dr$$

c) $di = \frac{dq}{I} = \sigma \omega r dr$

cc) campo magnético de una espira circular en su centro

$$dB = \frac{\mu_0 di}{2r}$$

$$dB = \frac{\mu_0 \sigma \omega}{2} dr$$

$$B = \frac{\mu_0 \sigma \omega R}{2}$$

ccc) Biot-Savart:

$$d\vec{B} = \frac{\mu_0 i}{4\pi} \frac{d\vec{l} \times \hat{d}}{d^2}$$

$$dB_y = \frac{\mu_0 i}{4\pi} \frac{dl}{d^2} \cos\theta$$

integrando

$$B_y = \frac{\mu_0 i 2\pi R \omega}{4\pi d^2}$$

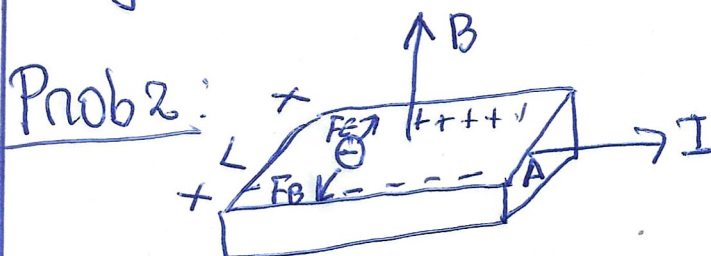
$$B_y = \frac{\mu_0 i r^2}{2 (z^2 + r^2)^{3/2}}$$

como $di = \sigma \omega r dr$

$$dB_y = \frac{\mu_0 \sigma \omega}{2} \frac{r^3 dr}{(z^2 + r^2)^{3/2}}$$

integrando $0 \rightarrow R$

$$B_y = \frac{\mu_0 \sigma \omega}{2} \left[\frac{2z^2 + R^2}{\sqrt{z^2 + R^2}} - 2z \right]$$



$$F_E = F_B$$

$$|E_H = v_d B| \quad \text{or} \quad |\Delta V_H = E_H L| \quad (2)$$

1 y 2 $V_d = \frac{\Delta V_H}{BL} = \text{reemplazando datos}$

$$v_d = 1.067 \times 10^{-4} = 1.07 \times 10^{-4} \frac{m}{s}$$

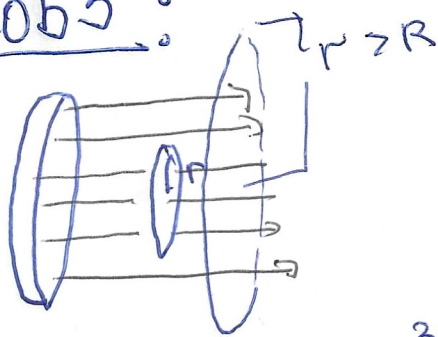
Como $I = n q v_d A = \frac{I}{A}$

$$\eta = \frac{I}{A q v_d}$$

reemplazando datos

$$\eta = 5.85 \times 10^{28} \frac{\text{Port}}{m^3}$$

Prob 3:



a) $\tau = RC = 2.0 \times 10^{-3} \text{ s}$

b) Proceso de carga del Condensador

$$q = Q_0 (1 - e^{-t/\tau})$$

Campo Electrico del Condensador

$$E = \frac{q}{\pi \epsilon_0 R^2} = \frac{Q_0}{\pi \epsilon_0 R^2} (1 - e^{-t/\tau})$$

flujo

$$\Phi = \frac{Q_0}{\pi \epsilon_0 R^2} (1 - e^{-t/\tau}) \text{ A}$$

$$\left| \frac{d\Phi}{dt} = \frac{Q_0}{\pi \epsilon_0 R^2} \frac{A}{\tau} \right|$$

c) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\Phi}{dt}$

$$A = \pi r^2$$

$$\left\{ B = \frac{\mu_0 Q_0}{2\pi R^2} \frac{r}{\tau} e^{-t/\tau} \right.$$

$$B = (2.50 \times 10^{-11}) r e^{-t/2000}$$

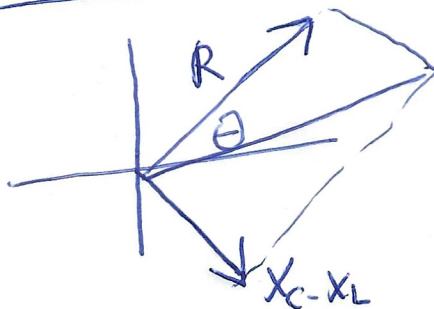
cc) $A = \pi R^2 \quad r > R$

$$\left\{ B = \frac{\mu_0 Q_0}{2\pi \tau} \frac{1}{r} e^{-t/\tau} \right.$$

$$B = \frac{(10^{-12})}{r} e^{-t/2000}$$

Prob 4 de los datos

$$\Theta = -54^\circ$$



c) $X_L = X_C + R \tan(\theta)$

$$X_L = 102.25 \Omega \quad \theta = -54^\circ$$

$$110 \Omega \quad \theta = -53^\circ$$

cc) $P_m = I_{ef} V_{ef} \cos \phi = I_{ef}^2 R$

$$I_{ef} = \sqrt{\frac{P_m}{R}} = \frac{\sqrt{7}}{3} = 0.882 \text{ A}$$

ccc) $V_{ef} = \frac{P_m}{I_{ef} \cos \phi}$

$$\left\{ \begin{array}{l} V_{ef} = 270 \text{ V} \quad \text{si } \theta = -54^\circ \\ V_{ef} = 265 \text{ V} \quad \text{si } \theta = -53^\circ \end{array} \right.$$